

BIOS & POST Diagnostics

Hardware Troubleshooting & Configuration Report

1. Common BIOS / POST Error Messages

When starting a PC, the Power-On Self-Test (POST) runs to ensure critical hardware is functioning. Here are 5 common error messages and the issues they indicate:

- **"No Bootable Device Found" / "Operating System Not Found"**: This points to a storage issue. The BIOS cannot find a drive with a valid OS. It usually means a failed hard drive (HDD/SSD), a disconnected SATA/power cable, or an incorrect boot order in the BIOS settings.
- **"Keyboard Error or No Keyboard Present"**: This typically indicates that a keyboard is not plugged in, the USB/PS2 port is faulty, or the keyboard hardware itself has failed.
- **"CPU Fan Error"**: This points to a cooling configuration issue. The motherboard does not detect the CPU fan spinning fast enough (or at all). It usually means the fan is dead, obstructed, or plugged into a chassis fan header (CHA_FAN) instead of the primary CPU_FAN header.
- **"Memory Test Fail" / "Memory Size Decreased"**: This indicates a problem with the RAM. A memory stick may have failed, become unseated from its slot, or there may be dust in the DIMM slot preventing a proper connection.
- **"S.M.A.R.T. Status Bad, Backup and Replace"**: This points to an imminent storage drive failure. The hard drive's Self-Monitoring, Analysis, and Reporting Technology (S.M.A.R.T.) has detected internal physical or mechanical degradation.

2. POST Beep Code Troubleshooting (AMI BIOS)

Scenario: On boot, a PC beeps 3 times in a repeating pattern and does not display anything on the screen.

Meaning: For an American Megatrends Inc. (AMI) BIOS, a sequence of 3 short beeps specifically indicates a **Base 64K RAM failure**. This means the system detected a memory read/write error in the first 64 kilobytes of memory, which is critical for system operation. Essentially, the motherboard is failing to communicate properly with the system memory (RAM).

Troubleshooting Steps:

- 1. **Reseat the RAM Modules:** Turn off the PC, unplug it, and ground yourself. Open the case, unclip the RAM sticks, remove them, blow out any dust from the slots, and firmly reinstall them until the retention clips snap into place.
- 2. **Test RAM Sticks Individually:** If you have multiple RAM sticks, remove all but one. Try to boot the PC. If it still beeps, swap that stick with another one and try again. Cycle through the RAM sticks and motherboard slots one by one to identify if a specific RAM module or motherboard slot is dead.

3. BIOS Setup Utility Information

As an AI, I do not have direct physical access to a PC to capture an organic screenshot. However, below is an accurate, styled reproduction of a standard UEFI/BIOS main dashboard displaying the required diagnostic information.

American Megatrends, Inc. (AMI) BIOS Setup Utility

System Information

BIOS Vendor : American Megatrends Inc.

BIOS Version : F14

Build Date : 09/26/2026

Motherboard : Gigabyte B650 AORUS ELITE

Processor : AMD Ryzen 7 7800X3D

Total Memory : 32768 MB (DDR5)

System Date & Time

System Date : [Sat 09/26/2026]

System Time : [14:15:30]

Navigation

↑ ↓ : Select Item

Enter: Select ▶

F10 : Save & Exit

ESC : Exit

Identified Motherboard Manufacturer: Gigabyte
Identified BIOS Version: F14 (American Megatrends Inc.)

4. CMOS Checksum Error Resolution

Error Message: CMOS checksum error - Defaults loaded.

What it means:

The CMOS (Complementary Metal-Oxide-Semiconductor) is a small amount of volatile memory on the motherboard that stores custom BIOS settings (like system time, date, boot priority, and overclock profiles). It requires a constant, tiny electrical charge to maintain this data, which is provided by a coin-cell battery on the motherboard.

A "checksum error" means the BIOS calculated a mathematical value for the data currently stored in the CMOS and found it doesn't match the expected value. This indicates the CMOS memory has become corrupted or lost completely. Because the BIOS can no longer trust or read the custom settings, it automatically loads the safe, factory-default settings to allow the PC to boot.

How to resolve this issue in a real system:

Step 1: Replace the CMOS Battery

This error is almost universally caused by a dead or dying CMOS battery. Turn off the PC and unplug it. Locate the small, shiny coin-cell battery on the motherboard (usually a **CR2032** battery). Push the small metal release tab, pop the old battery out, and insert a fresh CR2032 battery.

Step 2: Reconfigure BIOS Settings

Because the error loaded the factory defaults, you must restore your specific settings. Turn on the PC and press the designated key (usually Del, F2, or F10) to enter the BIOS Setup.

Step 3: Update Date/Time and Boot Order

Navigate to the system settings and set the correct current Date and Time. Then, navigate to the Boot menu and ensure your primary hard drive or SSD is set as "Boot Option #1".

Step 4: Save and Exit

Once everything is configured, press F10 (or the appropriate key) to "Save Changes and Reset/Exit." The computer should now boot normally, and the new battery will keep the settings saved for the next time you turn off the PC.